## SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

**Department of Computer Science and Engineering**

# CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

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| **Course Code:** | CSA1024 | **Course Title:** | Software Engineering |
| **CO Assessed:** | CO1 | **Assessment:** | Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop** |
| **Weightage:** | 20% | **Total Marks:** | 25 |
| **CO1 Statement:** | *Apply advanced methodologies and agile project management techniques to manage software projects effectively* | | |
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## Code of Conduct

*I **Dinesh V (192524309)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: **Dinesh V (192524309)**

## Annexure A – Agile Sprint Planning & User Story Workshop

Executive Summary

Pharmaceutical manufacturing depends on carefully controlled environmental conditions. Temperature, relative humidity, differential pressure, airborne particle concentration, air-flow velocity and door status can influence product quality, contamination control and the validity of a manufacturing batch. When these conditions are recorded manually or monitored by disconnected systems, a deviation may be detected late, supporting evidence may be incomplete, and teams at different facilities may not have the same view of the situation.

This assignment applies Agile software engineering practices to the proposed Pharmaceutical Manufacturing Environmental Monitoring Platform. The platform is envisioned as a distributed, multi-facility solution that receives telemetry from Internet of Things (IoT) sensors, evaluates measurements against approved product- and zone-specific limits, identifies excursions and trends, notifies Quality Assurance (QA), operations and facility teams, and maintains traceable records for review. It is also intended to support links with electronic batch records and Corrective and Preventive Action (CAPA) workflows.

The report follows the seven activities specified in the assessment brief: project vision and stakeholders; epics and user stories; acceptance criteria; product backlog prioritisation; first sprint planning and task breakdown; story-point estimation; and sprint goal and expected deliverables. User stories are written from a user's perspective, and the acceptance criteria use measurable Given–When–Then statements. A relative Fibonacci-style estimate is used to make uncertainty and implementation effort visible. The first sprint focuses on a dependable vertical slice: secure ingestion of simulated or test telemetry, configurable limits, basic excursion detection, and a traceable alert record. This scope is deliberately smaller than the complete production platform, because a real regulated deployment would require further validation, security review, qualification, integration testing and operational approval.

The proposed plan uses a two-week sprint as a planning assumption. The backlog, story points and schedule are illustrative estimates for this coursework exercise, not measured delivery commitments. Regulatory references are treated as design considerations: software features can support compliance readiness, but a system cannot be declared compliant solely because it contains an audit log or electronic signature screen.

# 1. Problem Statement Analysis

### 1.1 Industry context

Pharmaceutical and biotechnology facilities manufacture products in environments where air quality and physical conditions must be controlled according to the applicable process, facility design and approved procedures. A cleanroom, isolator or warehouse may have different acceptable operating ranges because the activities and materials handled in each area are not identical. For that reason, a useful monitoring system cannot rely on one universal threshold for every sensor and location.

The problem statement identifies six key telemetry categories: temperature, relative humidity, differential pressure, particle counts, air-flow velocity and door status. The platform must collect these measurements continuously, compare them with the limits configured for a product and zone, recognise an excursion, assess potential product impact, and make the event visible to the teams responsible for response and quality decisions.

### 1.2 Current-state challenges

| Challenge | Why it matters | Software response proposed |
|---|---|---|
| Manual or disconnected monitoring | Data can be delayed, duplicated or difficult to compare across rooms and sites. | Use a standard telemetry interface and central event pipeline. |
| Late detection of deviations | A prolonged excursion may affect product or process conditions before a person notices it. | Evaluate incoming readings promptly and create alerts when approved rules are met. |
| Different zone requirements | A warehouse, cleanroom and isolator may require different limits. | Maintain versioned, authorised product- and zone-specific configurations. |
| Incomplete investigation evidence | QA may need to reconstruct the event timeline, readings, acknowledgements and decisions. | Preserve timestamped events, rule versions, acknowledgements and audit history. |
| Siloed facilities | Central teams may lack a consistent view of site-level conditions. | Provide site-aware dashboards and role-based access across facilities. |
| Disconnected quality workflows | An excursion may need assessment, batch linkage and a CAPA record. | Provide controlled interfaces to electronic batch records and CAPA systems. |

### 1.3 Proposed system behaviour

1. Sensors or gateway devices publish readings with a sensor identifier, facility, zone, parameter, value, unit and timestamp.
2. The ingestion service validates the message structure, source identity, unit and timestamp, and rejects or quarantines malformed data.
3. The rules service obtains the currently approved threshold configuration for the relevant zone and product context.
4. The event engine evaluates the reading, including any configured persistence, dead-band or trend rules that have been approved by the process owner.
5. When a rule is breached, the system creates a traceable excursion event and assigns severity according to the approved configuration.
6. The notification service routes the event to the appropriate QA, operations and facilities recipients and records delivery or acknowledgement status.
7. The dashboard displays current conditions, alert state and historical trends. Authorised users can review the event and record their actions.

### 1.4 Software engineering interpretation

The task is not simply to display sensor values on a web page. It is a distributed software system with real-time data handling, configurable business rules, role-based permissions, integrations, data retention, auditability and availability

requirements. It also operates in a high-consequence context: incorrect limits, lost data or misleading alerts could cause users to make poor decisions. Agile delivery should therefore combine incremental development with disciplined requirements traceability, verification, change control and documented review.

The assessment brief asks for an Agile Sprint Planning and User Story Workshop, so the main deliverable is a planning artefact rather than a claim that a validated production system has already been built. Any values, estimates and sprint durations below are proposed assumptions for the coursework and should be adjusted if the instructor provides different constraints.

## 2. Project Vision, Objectives and Scope

### 2.1 Project vision

To provide pharmaceutical manufacturing teams with a dependable, secure and traceable environmental monitoring platform that turns continuous sensor readings into timely, actionable information, supports consistent deviation response across multiple facilities, and helps quality teams prepare reliable evidence for batch review and regulatory inspection.

### 2.2 Product vision statement

For QA professionals, production operators, facility engineers and site managers who need to maintain controlled manufacturing environments, the Pharmaceutical Manufacturing Environmental Monitoring Platform is a multi-site monitoring and event-management solution that collects environmental telemetry, checks approved limits, raises traceable alerts and supports investigation. Unlike paper-based logs or isolated monitoring tools, it aims to provide a consistent view of conditions and event history while retaining site-specific rules and access controls.

### 2.3 Objectives and measurable indicators

| Objective | Proposed indicator | Measurement approach |
|---|---|---|
| Collect environmental telemetry continuously | Percentage of valid test messages accepted by ingestion | Compare sent test messages with accepted, rejected and quarantined counts. |
| Detect excursions promptly | Time from receipt of a violating reading to event creation | Timestamp test events and measure processing latency. |
| Route alerts to the right teams | Percentage of test alerts delivered to configured recipients | Use test notification channels and delivery logs. |
| Preserve traceability | Percentage of key event actions represented in the audit trail | Check event creation, acknowledgement, configuration changes and closure actions. |
| Support historical analysis | Ability to retrieve a defined time window by facility, zone and parameter | Run functional queries over seeded test data. |
| Support multi-facility use | Authorised users can access permitted sites and cannot access prohibited sites | Perform role and tenant-boundary tests. |
| Improve inspection readiness | Report contains event details, timestamps, rule version and action history | Review generated sample report against an agreed checklist. |

These indicators are proposed acceptance measures for software development. They are not promises of a particular manufacturing outcome, and the assignment does not provide operational data from which reductions in batch rejection, recalls or cost can be calculated.

### 2.4 Scope

**In scope for the proposed product**

- Telemetry ingestion for temperature, relative humidity, differential pressure, particle counts, air-flow velocity and door status.
- Facility, building, room/zone, device and parameter metadata.
- Configurable product- and zone-specific threshold rules with authorised changes and version history.
- Excursion creation, severity, acknowledgement, assignment and status tracking.
- Role-based dashboards, event history and basic trend visualisation.
- Notifications to configured QA, operations and facilities recipients.
- Audit-trail events for relevant user and system actions.
- Exportable reports and integration points for electronic batch records and CAPA systems.
- Architecture that can be extended to additional facilities and devices.

**Out of scope for the first sprint**

- Connecting every production sensor and every facility in the organisation.
- Replacing the organisation's quality management system, electronic batch record system or validated control systems.
- Automatically deciding that a batch is safe or unsafe without authorised quality review.
- Claiming full regulatory compliance before documented assessment, validation and approval.
- Deploying to a live manufacturing environment without security, qualification, change-control and operational approvals.

### 2.5 Assumptions and constraints

| Type | Planning assumption |
|---|---|
| Sprint length | Two weeks for the initial sprint; this is an assumption for planning. |
| Team | A small cross-functional Scrum-style team with software, testing, product and domain input. |
| Sensor access | Sprint 1 can use simulated or test telemetry if physical sensors are not available. |
| Threshold values | Only approved example limits supplied by a domain owner may be used; no universal GMP values are invented in this report. |
| Regulatory use | The coursework prototype demonstrates compliance-supporting controls, not certified or validated production readiness. |
| Integration | Batch-record and CAPA connections begin with documented interface contracts or stubs if external systems are unavailable. |
| Data protection | Test data should be synthetic; production records and credentials must not be placed in development examples. |

## 3. Stakeholder Identification and Responsibilities

Stakeholders are people or organisations that influence the product, use it, provide its data, approve its operation or are affected by its outputs. In a regulated manufacturing setting, stakeholder identification is especially important because a technical feature can affect a quality process. The product owner should not decide technical acceptance criteria in isolation when those criteria depend on QA procedures or facility engineering.

| Stakeholder | Main interests / needs | Responsibilities and involvement |
|---|---|---|
| Quality Assurance (QA) | Reliable deviation records, clear severity, traceable review and evidence. | Approves quality-related workflows and threshold governance; reviews excursion records; confirms whether event handling meets procedures. |
| Production / Operations | Current room status, timely notifications and practical response instructions. | Acknowledges assigned alerts, follows approved operating procedures and records actions. |
| Facility / HVAC Engineering | Pressure, airflow, equipment status and sensor health. | Maintains sensors and building systems; investigates facility-related excursions; supports calibration and device diagnostics. |
| Microbiology / Environmental Monitoring team | Particle and environmental trends, monitoring context and investigation support. | Reviews monitoring trends and supplies domain rules where this falls within its responsibility. |
| Site Quality Head / Quality Unit | Consistent quality governance and controlled multi-site operation. | Approves site procedures, escalation rules and quality-related release or disposition decisions. |
| Manufacturing / Batch-record team | Correct association between environmental events and batches. | Defines batch identifiers, integration requirements and rules for linking events to batch records. |
| IT / Platform Operations | Availability, maintainability, backup, monitoring and support. | Operates the platform, manages environments, deploys approved releases and responds to incidents. |
| Information Security | Confidentiality, integrity, identity controls and secure transmission. | Reviews threat model, authentication, authorisation, encryption, logging and vulnerability remediation. |
| Regulatory / Compliance specialists | Evidence aligned with applicable requirements and intended use. | Advise on applicability and interpretation of relevant regulations and guidance; review compliance documentation. |
| Data integrity / Validation team | Requirements traceability, test evidence, controlled changes and reliable records. | Defines validation strategy, reviews test evidence and supports release qualification. |
| Sensor / IoT vendors | Clear device protocols and integration expectations. | Provide device specifications, calibration information, supported interfaces and maintenance guidance. |
| Product Owner | Maximum value delivered in the correct order. | Owns product vision, orders the backlog, clarifies stories and accepts completed work. |
| Scrum Master / Agile facilitator | Healthy process, transparency and removal of impediments. | Facilitates planning, daily coordination, review and retrospective; helps the team resolve blockers. |
| Development team | Clear requirements, manageable tasks and feedback. | Designs, implements, tests, documents and integrates product increments. |
| Auditors / inspectors | Understandable, retrievable and trustworthy evidence. | Review records and system controls within their authorised inspection remit; they are not day-to-day product operators. |

### 3.1 Stakeholder engagement plan

| Activity | Participants | Frequency / output |
|---|---|---|
| Requirements workshop | Product Owner, QA, operations, facilities, security, developers | Before backlog refinement; agreed terms, risks and initial story candidates. |
| Backlog refinement | Product Owner and delivery team, with domain experts when needed | At least once per sprint; stories clarified, dependencies exposed and acceptance criteria improved. |
| Sprint Planning | Whole Scrum team | Start of sprint; sprint goal, selected stories and task plan. |
| Daily Scrum / daily coordination | Developers; facilitator supports as needed | Daily; inspect progress toward sprint goal and adapt the plan. |
| Sprint Review | Team and relevant stakeholders | End of sprint; demonstrate working increment and gather feedback. |
| Sprint Retrospective | Scrum team | End of sprint; agree one or more process improvements. |
| Quality / security review | QA, validation, security and technical representatives | At defined design and release gates; record risks, decisions and required evidence. |

### 3.2 Stakeholder communication principles

Communication should be timely, role-appropriate and recorded where it affects a quality decision. An alert notification is not itself proof that an excursion has been investigated. The application should distinguish the time a condition occurred, the time the platform received it, the time an alert was generated, the time it was delivered, and the time a person acknowledged it. This separation makes later review more meaningful and helps identify network delays or gaps in response.

## 4. Agile Approach and Working Agreements

### 4.1 Why Agile is appropriate

The platform has several interacting requirements and interfaces. Some requirements will become clearer after stakeholders see a working dashboard or a sample excursion record. Agile delivery makes it possible to produce small increments, obtain feedback early and revise the backlog as new information emerges. Rather than attempting to build all sensors, reports and integrations at once, the team can first establish a dependable path from telemetry to an alert, then extend it with trend analysis, batch linkage and multi-site administration.

Agile does not mean skipping documentation or testing. In this domain, the team should maintain requirements traceability, version control, review records, test evidence, controlled configuration and clear release decisions. The amount of documentation should be appropriate to intended use and risk. The Product Owner orders the backlog, while QA, security and validation specialists provide necessary domain oversight.

### 4.2 Suggested team and roles

| Role | Contribution to this project |
|---|---|
| Product Owner | Represents user and business needs; prioritises backlog; clarifies acceptance criteria; makes product-scope trade-offs. |
| Scrum Master / facilitator | Facilitates events, promotes transparency, identifies impediments and protects focus on the sprint goal. |

| | |
|---|---|
| Developers | Collectively design, implement, test, integrate and document the increment. The team may include frontend, backend, IoT, data and test skills. |
| QA / domain representative | Clarifies environmental event workflows, severity rules, quality impact assessment and review expectations. |
| Security / validation support | Reviews access controls, data integrity, audit history, change control and evidence needs. |

### 4.3 Working agreements

- Keep user stories focused on a user outcome rather than a particular technical implementation.
- Do not use unapproved threshold values as real manufacturing limits; mark all sample data as synthetic.
- Changes to threshold rules must be authorised, versioned and attributable to a user or approved system process.
- A story is not complete merely because the screen looks correct; relevant negative, boundary and permission tests must pass.
- No production credentials, personal data or genuine batch data should be committed to source control or included in screenshots.
- If an external interface is unavailable, use a clearly labelled test stub and document what remains unverified.
- Raise uncertainty early. A blocked story should be visible in the daily coordination rather than hidden until the sprint ends.
- The Sprint Review demonstrates the increment and gathers feedback; it does not replace formal validation or release approval.

### 4.4 High-level architecture as a planning aid

A proposed logical architecture helps the team divide work without locking the project into a specific vendor or technology. Sensor gateways send signed or authenticated telemetry to an ingestion API or message broker. Validation and normalisation services prepare the readings for the rules engine. The rules engine evaluates approved thresholds and produces events. A notification service distributes alerts, while an application API supports dashboards, event review, historical queries and report generation. Data storage separates operational telemetry from configuration and audit records where appropriate. An integration layer exchanges approved identifiers and event references with batch-record and CAPA systems.

| Logical component | Primary responsibility | Sprint 1 treatment |
|---|---|---|
| Sensor / gateway interface | Produce telemetry with identity, time, parameter, unit and value. | Use a simulator or test payloads if devices are unavailable. |
| Ingestion and validation | Authenticate source where supported; validate schema, timestamp and units. | Implement basic schema validation and rejection handling. |
| Threshold configuration | Store authorised limits and applicable zone/parameter context. | Use a small controlled test configuration with version identifier. |
| Excursion engine | Evaluate measurements and create events when approved rules are breached. | Implement basic upper/lower threshold checks for test cases. |
| Alert service | Route notifications and record delivery/acknowledgement state. | Create a test alert record and use a sandbox notification channel. |
| Dashboard / API | Show current readings and events to authorised users. | Provide a basic event/readings view or API response. |
| Audit and persistence | Preserve traceable records and relevant action history. | Persist event creation and key status changes in test storage. |
| Integration adapter | Link events to batch and CAPA systems. | Document interface contract; full integration deferred. |

## 5. Epics and User Stories

An epic is a broad capability that is too large to complete as one small backlog item. A user story expresses a useful outcome from the viewpoint of a person or system that needs it. The following epics translate the supplied problem statement into incremental product capabilities. Story IDs are included so that the backlog, acceptance criteria, sprint plan and tests can refer to the same item.

**5.1 Epic catalogue**

| Epic ID / name | Purpose | Business value |
|---|---|---|
| E1 – Telemetry collection | Receive and validate readings from sensors and gateways. | Creates the data foundation for monitoring. |
| E2 – Zone and threshold management | Maintain facility, zone, parameter and approved limit configuration. | Ensures measurements are evaluated in the correct context. |
| E3 – Excursion detection | Identify readings outside configured limits and create event records. | Enables faster and more consistent deviation awareness. |
| E4 – Alerts and response | Notify relevant teams, acknowledge events and track response. | Reduces the chance that a detected event is overlooked. |
| E5 – Product impact and batch linkage | Associate events with products, batches and quality review workflows. | Supports documented product-impact assessment. |
| E6 – Audit trail and compliance reporting | Record key actions and generate reviewable reports. | Improves traceability and inspection preparation. |
| E7 – Historical analytics | Display trends and summaries over selected periods. | Helps identify recurring conditions and deterioration. |
| E8 – Multi-facility access and resilience | Support multiple sites, permissions, availability and recovery. | Supports growth while protecting site boundaries. |

**5.2 User stories**

| Story ID / Epic | User story | Priority | Points |
|---|---|---|---|
| US-01 / E1 | As a facility engineer, I want the platform to receive timestamped sensor readings, so that environmental conditions can be monitored without relying only on manual transcription. | High | 5 |
| US-02 / E1 | As a platform operator, I want invalid or incomplete telemetry to be rejected or quarantined with a reason, so that bad data does not silently affect quality decisions. | High | 5 |
| US-03 / E2 | As an authorised QA user, I want to configure and version approved limits by facility, zone and parameter, so that each reading is evaluated against the correct requirements. | High | 8 |
| US-04 / E3 | As a QA specialist, I want the system to create an excursion when a valid reading violates an active rule, so that potential deviations are visible promptly. | Highest | 8 |
| US-05 / E4 | As an operations user, I want an alert to identify the zone, parameter, reading, limit and event time, so that I can understand what needs attention. | Highest | 5 |
| US-06 / E4 | As a responsible user, I want to acknowledge and update the status of an assigned event, so that the response history is visible to the team. | High | 5 |
| US-07 / E6 | As a QA reviewer, I want important event and configuration actions recorded with actor and timestamp, so that the history can be reconstructed during an investigation. | High | 8 |
| US-08 / E7 | As a facility engineer, I want to view historical readings by site, zone, parameter and time range, so that recurring trends can be investigated. | Medium | 8 |
| US-09 / E5 | As a QA reviewer, I want to link an excursion to a relevant batch record, so that product-impact assessment has the necessary context. | High | 8 |
| US-10 / E5 | As a quality specialist, I want to refer an event to the CAPA workflow when required, so that corrective and preventive actions can be tracked in the quality system. | Medium | 8 |
| US-11 / E6 | As an auditor or authorised QA user, I want to generaport containing readings, events, rule versions and response history, so that I can review the evidence for a selected period. | High | 8 |
| US-12 / E8 | As a corporate quality manager, I want to view permitted summaries across facilities, so that I can compare site status without bypassing site-level access restrictions. | Medium | 8 |
| US-13 / E8 | As a security administrator, I want access to be based on assigned roles and facility scope, so that users can only view or change information they are authorised to handle. | High | 8 |
| US-14 / E8 | As an IT operator, I want the platform to monitor ingestion health and recover safely from a service interruption, so that gaps are visible and service can be restored in a controlled way. | High | 13 |
| US-15 / E2 | As a QA approver, I want threshold changes to require authorised approval and retain the prior version, so that changes cannot silently rewrite the basis of earlier decisions. | High | 8 |
| US-16 / E3 | As a QA specialist, I want to distinguish a single reading from a sustained excursion according to an approved rule, so that alert behaviour reflects the defined process rather than arbitrary code assumptions. | Medium | 8 |

### 5.3 Notes on story quality

The stories use the requested format: “As a [user], I want [feature], so that [benefit].” They describe user value, not just technical tasks. For example, “build an API endpoint” is a development task; the corresponding user story is about receiving validated telemetry so that monitoring is dependable. The team may create technical enabler tasks beneath a story, but the backlog should remain connected to the intended outcome.

Some stories are intentionally broad at this stage and need refinement before a sprint. In particular, batch linkage, CAPA integration, multi-site access and high availability depend on the interfaces and operating model of the organisation. They should not be estimated as if those details were already known. During backlog refinement, the Product Owner and relevant specialists should confirm data contracts, role definitions, response rules and the meaning of completion.

## 6. Acceptance Criteria

Acceptance criteria define observable conditions that must be satisfied for a user story to be accepted. They should be specific enough for developers and testers to agree on expected behaviour, including normal, invalid and boundary cases. The examples below are proposed for the coursework. Actual threshold values and response rules must come from approved site procedures and authorised domain owners.

### 6.1 US-01 – Receive timestamped sensor readings

- Given a valid telemetry message with device ID, facility, zone, parameter, unit, value and timestamp, when the ingestion service receives it, then the service stores or queues the reading and returns a success outcome.
- Given a message missing a required field, when it is submitted, then the platform rejects or quarantines it and records a validation reason.
- Given a reading with a timestamp outside the configured acceptance window, when it is received, then the platform applies the documented late-data policy and does not silently treat it as a current reading.
- Given the same message identifier is submitted twice, when duplicate handling is enabled, then the platform prevents unintended duplicate event processing and records the duplicate outcome.

### 6.2 US-02 – Validate and quarantine invalid telemetry

- Given a parameter expects a defined unit, when a message uses an unsupported unit, then the platform rejects it or converts it only through an explicitly configured and tested conversion.
- Given a sensor ID is not registered or authorised, when it submits data, then the platform does not accept the reading as trusted telemetry and records a security or validation event.
- Given a payload cannot be parsed, when ingestion processes it, then the system records the reason without exposing secrets or crashing the ingestion service.
- Given a quarantined message is reviewed, when an authorised user opens its details, then the original payload reference, receipt time and validation reason are visible according to access policy.

### 6.3 US-03 – Configure and version limits

- Given an authorised QA user creates a rule, when required fields are valid, then the platform saves the parameter, applicable facility/zone context, units, limits, effective period, status and version.
- Given a user without configuration permission attempts to change a limit, when the request is submitted, then the change is denied and the attempt is recorded.
- Given a new version is approved, when it becomes effective, then subsequent readings use the effective version while historical events retain the rule version used at the time.
- Given a lower limit is greater than or equal to an upper limit where the parameter requires a range, when the rule is saved, then validation prevents activation and explains the problem.

### 6.4 US-04 – Detect an excursion

- Given a valid reading is within the active approved limits, when the rule engine evaluates it, then no out-of-limit excursion is created for that reading.
- Given a valid reading is above the configured upper limit or below the configured lower limit, when the rule engine evaluates it, then an excursion record is created with the reading, limit, zone, parameter, timestamps and rule version.
- Given a reading belongs to a zone with no active approved rule, when it is received, then the platform marks it as unassessed or rule-missing and does not invent a threshold.
- Given an event is created, when it is stored, then it receives a unique event identifier and can be retrieved through the authorised event view.

### 6.5 US-05 – Send an informative alert

- Given a new excursion meets the configured alert policy, when the event is created, then a notification is queued for the configured recipient group.
- Given the alert is displayed, then it includes event ID, facility, zone, parameter, observed value, unit, applicable limit, event/receipt time and severity.
- Given a notification provider fails, when delivery is attempted, then the failure is logged and the platform follows the configured retry or escalation policy.
- Given a user lacks permission to view the zone, when they follow an alert link, then the platform denies access without revealing restricted details.

### 6.6 US-06 – Acknowledge an event

- Given an authorised user opens an active event, when they acknowledge it, then the event status, actor, timestamp and optional comment are recorded.
- Given an unauthorised user attempts to acknowledge an event, when the request is submitted, then the action is denied and an audit/security record is created where required.
- Given an event is already acknowledged, when another acknowledgement is attempted, then the system applies a defined status-transition rule and does not erase the original acknowledgement.
- Given an event is closed, when a user tries to reopen it, then the action is permitted only if the configured role and workflow allow it.

### 6.7 US-07 – Maintain an audit trail

- Given a significant event or configuration action occurs, when the action completes, then the audit record includes the actor or system identity, timestamp, action type and affected record reference.
- Given a user views an audit record, then the user can distinguish the event's original value from later workflow updates where applicable.
- Given a standard user attempts to alter or delete a protected audit entry, when the request is made, then the operation is denied.
- Given an audit export is generated, then it includes the selected period, filter criteria and generation time and is restricted to authorised users.

### 6.8 US-08 – View historical trends

- Given a user selects an authorised facility, zone, parameter and time range, when the query runs, then the platform returns matching readings in chronological order.
- Given the selected period contains no readings, then the interface displays a clear empty-state message rather than a misleading zero value.
- Given the data contains a known gap, then the trend view indicates missing data or an unavailable interval where that information is available.

- Given the user requests a large time range, then the platform applies documented limits or aggregation and communicates the selected resolution.

**6.9 US-09 and US-10 – Batch and CAPA linkage**

- Given a QA user has permission to link an event to a batch, when a valid batch identifier is selected, then the system records the link, actor, timestamp and source system reference.
- Given the batch identifier cannot be verified, then the platform clearly marks the link as unverified or rejects it according to the integration contract.
- Given a quality process requires CAPA referral, when an authorised user initiates the referral, then the platform sends the required event reference and approved metadata to the CAPA interface.
- Given the external system is unavailable, then the platform records the failed integration attempt and provides a retry or reconciliation path without falsely claiming success.

**6.10 US-11 – Generate an audit-ready report**

- Given an authorised user chooses a facility and reporting period, when the report is generated, then it includes selected environmental data, excursions, applicable rule versions, acknowledgements and recorded actions.
- Given a report contains no events, then the report states that no matching events were found for the selected filters rather than implying the monitoring system was offline.
- Given the report is exported, then its generation timestamp, reporting period, filters and user identity are recorded.
- Given the user is not authorised to access a facility, then the report does not include that facility's restricted data.

**6.11 Compliance-related acceptance boundary**

The platform may be designed to support electronic records, controlled access, audit trails, electronic signatures where required, record retention and validated workflows. Whether a particular regulation applies, and what controls are required, must be determined for the system's intended use and jurisdiction. An acceptance test can verify a specific control; it cannot by itself establish that the whole platform complies with every applicable regulatory requirement.

## 7. Product Backlog Prioritisation

### 7.1 Prioritisation method

The backlog is prioritised using a practical MoSCoW-style approach, informed by user value, risk reduction, dependencies and implementation uncertainty. “Must” items are needed for a credible initial monitoring slice; “Should” items add important response or traceability capability; “Could” items improve analysis or convenience; and “Later” items depend on more detailed interfaces or infrastructure. Priority does not mean that a lower-priority feature is unimportant. It means that the team should establish the foundational workflow before taking on more complex capabilities.

In this project, ingestion, valid configuration and excursion detection are foundational dependencies. Alerts are valuable only when the event data is reliable and the rule applied is known. Reporting and product-impact linkage also depend on traceable event identifiers and history. Multi-site analytics depends on reliable site metadata and access boundaries.

| Order | Story | Priority | Points | Reason / dependency |
|---|---|---|---|---|
| 1 | US-01 – Receive telemetry | Must | 5 | Foundation for all monitoring capabilities. |
| 2 | US-02 – Validate telemetry | Must | 5 | Prevents malformed data from silently entering processing. |
| 3 | US-03 – Configure limits | Must | 8 | Provides approved context for evaluation. |
| 4 | US-04 – Detect excursion | Must | 8 | Core value: turn readings into actionable events. |
| 5 | US-05 – Informative alert | Must | 5 | Makes detected events visible to responsible teams. |
| 6 | US-06 – Acknowledge event | Should | 5 | Adds a basic response workflow and ownership. |
| 7 | US-07 – Audit trail | Must/Should | 8 | Traceability is central; initial event history is prioritised, with deeper controls refined. |
| 8 | US-13 – Role-based access | Must | 8 | Security and facility separation underpin safe access. |
| 9 | US-15 – Approved rule changes | Should | 8 | Important for governance; may be split into a later increment. |
| 10 | US-08 – Historical trends | Should | 8 | Useful after reliable time-series data exists. |
| 11 | US-11 – Audit report | Should | 8 | Depends on event and audit data structures. |
| 12 | US-09 – Batch linkage | Should | 8 | Requires batch identifier and integration contract. |
| 13 | US-10 – CAPA referral | Could/Should | 8 | Depends on external quality-system interface. |
| 14 | US-16 – Sustained excursion rules | Should | 8 | Requires approved persistence/dead-band semantics. |
| 15 | US-17 – Alert escalation | Could | 5 | Depends on alert status and response-time policy. |
| 16 | US-12 – Cross-site summary | Could | 8 | Depends on multi-site metadata and access controls. |
| 17 | US-18 – Repeated-excursion summary | Could | 8 | Depends on sufficient historical data. |
| 18 | US-14 – Recovery and health monitoring | Must for production; later increment | 13 | High value but broad; needs infrastructure and recovery design. |

### 7.2 Dependency logic

- US-01 and US-02 precede reliable rules evaluation because the system needs valid, contextualised readings.

- US-03 precedes US-04 because an excursion must be compared with a known active rule.
- US-04 precedes US-05 because the alert needs an event record and its details.
- US-04 and US-06 provide the event lifecycle that US-07 must record.
- US-01 and event persistence precede US-08 because trend queries need stored time-series data.
- US-07 and event data precede US-11 because a report must include reliable history and rule context.
- US-09 and US-10 depend on confirmed external identifiers, data contracts and error-handling behaviour.
- US-12 and US-14 require architecture, permissions, monitoring and operational decisions beyond a minimal prototype.

### 7.3 Prioritisation rationale

The initial backlog favours reducing the risk of incorrect or invisible monitoring decisions. A visually polished dashboard is not the first priority if the underlying data is not validated or if the platform cannot explain which rule generated an excursion. Similarly, a report should not be treated as audit-ready merely because it is downloadable; it must be reproducible, traceable to source records and protected by suitable access controls. These considerations make data correctness and event traceability higher priorities than optional visual refinements.

## Conclusion

The Pharmaceutical Manufacturing Environmental Monitoring Platform addresses a meaningful software engineering problem: environmental data must be collected consistently, interpreted against approved rules, turned into timely events and preserved in a form that supports investigation. A distributed design can improve visibility across facilities, but it also introduces responsibilities for data quality, access control, integration, availability and traceability. These responsibilities should be made explicit in the product backlog from the beginning.

This Agile planning document translates the problem statement into a vision, stakeholder map, epics, user stories, measurable acceptance criteria, a prioritised backlog, a first sprint plan, task breakdown, story-point estimates and a sprint goal. The proposed first sprint deliberately focuses on an end-to-end telemetry-to-alert path. This is a practical foundation because later capabilities—historical trends, batch linkage, CAPA workflows, audit reports and multi-site views—depend on reliable event data and well-defined configuration.

The estimates and schedule are illustrative and should be confirmed with the actual team. Similarly, the report identifies compliance-supporting design considerations without claiming that a prototype is automatically compliant with 21 CFR Part 11 or any other requirement. Production readiness requires appropriate regulatory interpretation, risk assessment, validation, security review and controlled operational approval.

In conclusion, Agile methods can help deliver the platform incrementally while maintaining transparency and stakeholder feedback. The most important measure of progress is not simply that a feature has been coded, but that the feature provides a clear user outcome, behaves predictably under test, preserves relevant evidence and is understood by the people who will rely on it.

**Rubrics for Calculation**

| Criteria | Excellent (5 Marks) | Good (3 Marks) | Satisfactory (2 Marks) | Needs Improvement (1 Mark) |
|---|---|---|---|---|
| **Project Vision & Stakeholder Identification (30%)** | Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified. | Vision is clear with minor omissions. Most stakeholders are correctly identified. | Vision is partially defined. Some stakeholders are missing or incorrectly identified. | Vision is unclear or incomplete. Stakeholders are poorly identified or missing. |
| **Epics & User Stories (25%)** | Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements. | Epics and user stories are mostly complete with minor formatting or requirement issues. | User stories are partially complete with limited Agile structure. | User stories are incomplete, unclear, or do not follow Agile practices. |
| **Acceptance Criteria (20%)** | Acceptance criteria are specific, measurable, testable, and aligned with all user stories. | Acceptance criteria are mostly clear and testable with minor gaps. | Acceptance criteria are partially defined and lack clarity or completeness. | Acceptance criteria are missing or not aligned with user stories. |
| **Product Backlog Prioritization & Sprint Planning (15%)** | Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete. | Backlog prioritization and sprint planning are mostly appropriate with minor issues. | Prioritization or sprint planning is partially complete with limited justification. | Backlog prioritization and sprint planning are poorly organized or incomplete. |
| **Story Point Estimation, Sprint Goal & Presentation (10%)** | Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized. | Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear. | Estimation or sprint goal lacks justification. Presentation is adequate but incomplete. | Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete. |

| Criteria | **Mark** |
|---|---|
| Project Vision & Stakeholder Identification **(30%)** | /5 |
| Epics & User Stories **(25%)** | /5 |
| Acceptance Criteria **(20%)** | /5 |
| Product Backlog Prioritization & Sprint Planning **(15%)** | /5 |